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Progress Report 2 on

Power System Assessment of Dynamic Line Rating for Alberta Transmission Utilities

Course Name: Graduate Project in Electrical Engineering

Course Title: ENEL **678**

Submitted By,

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Progress Made So Far

Project Deliverables:

1. **Deliverable 1: DLR Hardware Prototype:** The Arduino-based sensing system remained fully operational during this phase, providing stable measurements of temperature, wind speed, and light intensity for DLR computation. **Completion: 100%**

Description: Stable environmental sensing system supporting real-time DLR calculation and simulation input.

2. **Deliverable 2: IEEE 738 DLR Model:** The IEEE 738 heat-balance model was successfully implemented in Python to compute dynamic ampacity using logged environmental data. **Completion: 100%**

Description: Dynamic ampacity calculation using IEEE 738 for integration with PSSE-based analysis.

3. **Deliverable 3: PSSE Power Flow Workflow:** A PSSE-based workflow was developed to import DLR data through Python scripting and evaluate load flow and N-1 contingency performance in an Alberta transmission network. **Completion: 100%**

Description: Alberta base case was solved in PSSE, and DLR values were converted from A to MVA and applied to selected lines using the PSSE API. Initial SLR vs DLR load-flow validation was completed, and the workflow is prepared for contingency studies.

4. **Deliverable 4: Grid Impact and Comparative Analysis:** The DLR model was integrated with PSSE to update transmission line ratings and evaluate system performance. Comparative analysis across selected 138 kV lines shows reduced loading under DLR conditions. **Completion: 80%**

Description: Preliminary validation of DLR impact through multi-line comparison and PSSE-based simulation.

Methodology

The project methodology followed an end-to-end validation workflow shown below:

Sensors → Arduino → Python IEEE 738 model → DLR ampacity → A-to-MVA conversion → PSSE branch update → Load flow comparison → N-1 contingency test → Final results

This methodology remains aligned with the approved Design Roadmap by preserving the planned sequence of sensing, DLR computation, PSSE integration, and contingency-based system validation.

1. **Environmental Data Acquisition:** Measured ambient temperature, wind speed, and solar irradiance using the Arduino-based sensor system, then filtered and logged the data for processing.
2. **DLR Computation:** The logged dataset was processed in Python using the IEEE 738 heat-balance model to compute conductor dynamic ampacity.
3. **PSSE Integration:** The Alberta base case was solved, candidate 138 kV lines were selected, computed ampacity was converted from A to MVA, and branch ratings were updated.
4. **Base Case vs DLR Case Comparison:** Converted computed ampacity values from A to MVA and updated selected branch ratings in PSSE using Python scripting, while limiting DLR to 130% of the static rating for simulation stability.

Line	Voltage (kV)	Base RATEA (MVA)	DLR RATEA (MVA)	Base Loading (%)	DLR Loading (%)
1303–1304 ckt 05	138	157.17	204.32	74.84	57.57

1241–1242 ckt 15	138	160.04	208.05	70.33	54.10
1164–1160 ckt 32	138	109.30	142.09	65.60	50.46

The actual power flow remained approximately unchanged, confirming that the reduction in loading is due to increased thermal capacity rather than changes in system dispatch.

5. **N-1 Contingency:** An N-1 contingency analysis was performed by removing line 1241–1242 and observing the response of monitored line 1303–1304 in both the base and DLR cases. Under normal operating conditions, loading on the monitored line decreased from 74.84% to 57.57% after applying DLR. This indicates that the increased line rating provided additional thermal headroom and reduced network stress. Early contingency runs showed convergence challenges under stressed conditions, so a capped and validated DLR rating was used for stable analysis. These results indicate that DLR can improve reliability, but only within practical operational limits.

Deviations and Justification:

1. A validated instructor-provided PSSE base case was used instead of manual network modeling to reduce development time and ensure a realistic simulation environment.
2. Offline CSV replay was used in place of continuous live streaming to enable repeatable validation and easier debugging.
3. DLR values were capped at 30% above static ratings to avoid unrealistic operating conditions and maintain power-flow convergence.

Challenges Encountered

Challenge	Description	Impact	Mitigation
Technical	Unit mismatch between DLR output (A) and PSSE input (MVA)	Incorrect branch rating updates	Converted ampacity to MVA using line voltage before updating PSSE
Technical	Dynamic update of PSSE branch ratings	Difficulty modifying branch limits through Python	Developed and verified psspy-based rating update scripts
Simulation	Numerical instability at high DLR values	Convergence issues during load flow and contingency studies	Limited DLR increase to 30% above static rating

Roles & Responsibilities

Responsible Person	Roles & Responsibilities
Ahmed Mashook Iqbal	Supervised project coordination and validated grid impact findings against project objectives.
G K Md Muttakin	Executed PSSE integration, branch rating updates, and N-1 contingency simulations for selected transmission lines.

Eisa Kanhoush	Prepared appendix materials and supported formatting, result presentation, and document refinement.
Monjurul Haque	Supported interpretation of DLR-based power flow improvements and comparative analysis.
Asif Rahman	Compiled simulation results, prepared technical write-up, and developed report tables and summaries.

List of Planned Work for the Following Month

Planned Tasks	Deliverables	Timeline
Result visualization and performance analysis	Generation of graphs (ampacity vs time, line loading comparison, % improvement) and summary tables	March 29
Preparation of project presentation	Slide deck summarizing system design, implementation, results, and key findings	April 5
Contingency result refinement	Final interpretation of non-convergent N-1 scenarios and validated discussion	April 7
Final report preparation and documentation	Complete technical report including methodology, results, discussion, and conclusions	April 9

Updated Risk Assessments

Risk	Impact	Likelihood	Mitigation	Status
Non-convergence during contingency simulations	Invalid load-flow results	Low	Limit DLR to 30% above static rating	Controlled
Overestimation of DLR under extreme conditions	Unrealistic performance improvement	Low–Medium	Use bounded limits and compare with SLR baseline	Controlled
Data synchronization between sensor input and simulation	Inconsistent or non-repeatable validation results	Medium	Use of CSV-based offline data replay for controlled testing	Mitigated

Budget Updates

The total expenditure on our hardware project is \$118 (CAD), which is within the original estimated budget and no major cost deviation has occurred. The breakdown is given in the appendix.

Conclusion

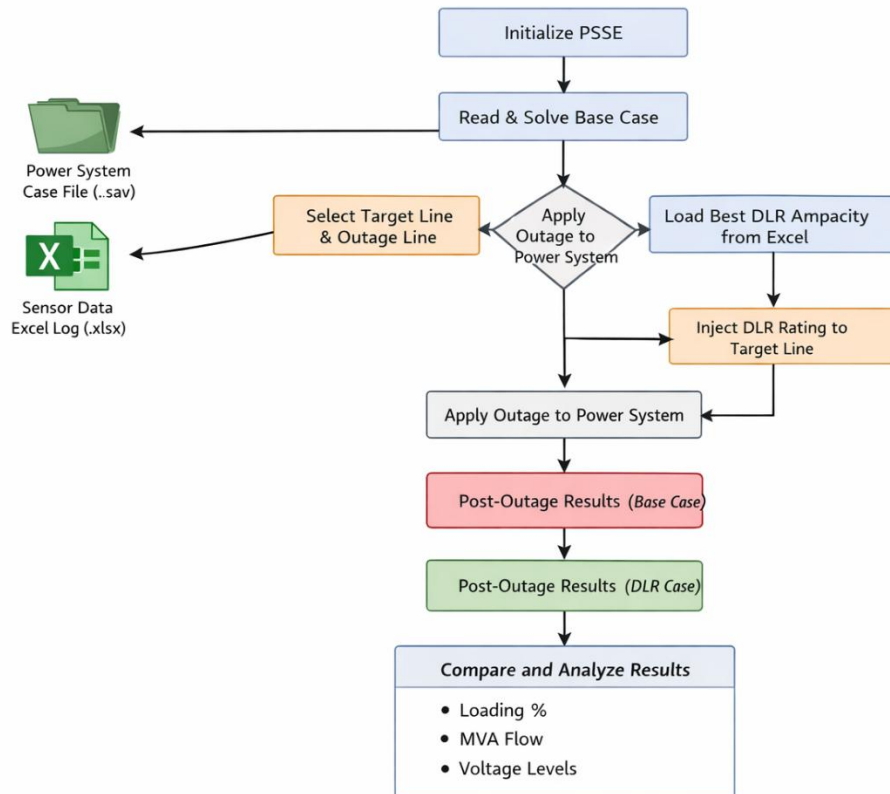
The project successfully integrated Arduino-based sensing, IEEE 738 DLR computation, and PSSE simulation into a complete validation workflow. Results show increased branch ratings and reduced loading compared to static ratings, while preliminary N-1 testing demonstrates the feasibility of contingency-based reliability assessment. The remaining work focuses on interpreting non-convergent contingency cases, final visualization, and documentation.

Appendix

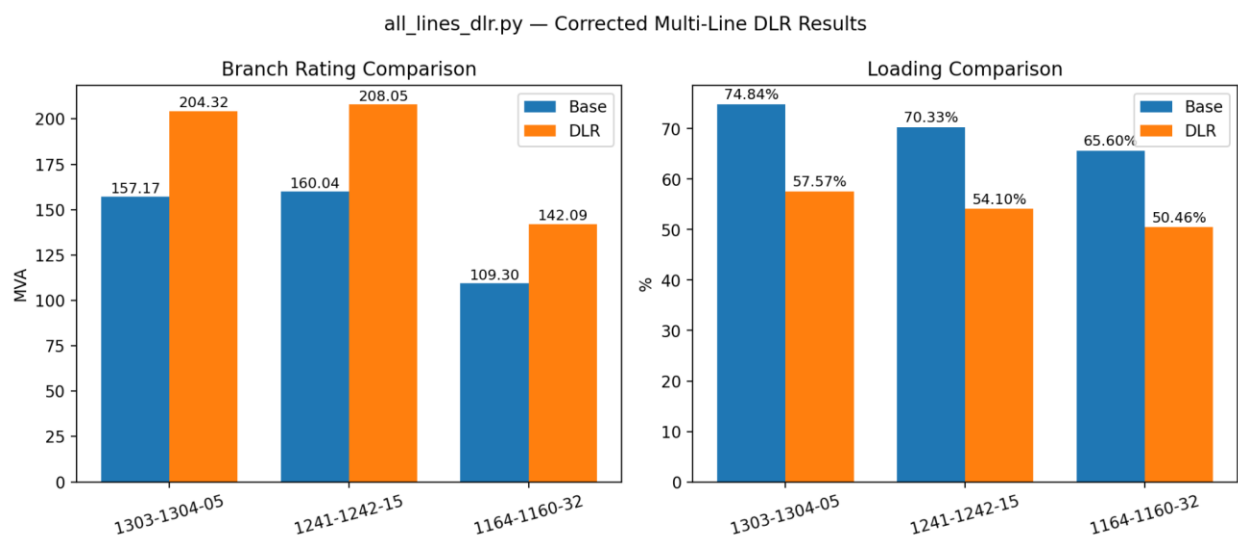
Appendix A: Budget Breakdown

Equipment Name	Model	Quantity	Cost (CAD)
Temperature Sensor	TT-PT-1000B-2050-11-AUNI-ND	1	\$ 4.79
Wall Mount Adapter	1939-WR9HD1333CCP-F(R6B)-ND	1	\$ 9
Wind Sensor	1528-1328-ND	1	\$ 69.3
Light Sensor	5032-333046-ND	1	\$ 4.95
Arduino Kit	Arduino UNO	1	\$ 0 (Makerspace)
Resistor Kit	1568-10969-ND	1	\$ 15.24
Film Capacitor	495-B32529C0104J289CT-ND	2	\$ 0.86
Digital Display	LCM1602C	1	\$ 0
Jumper Wire M TO M	1528-1967-ND	1	\$ 2.82
Jumper Wire M TO F	1528-1964-ND	1	\$ 2.82
Analog Sensor	LM35DT/NOPB-ND	1	\$ 4.1

Appendix B: DLR Workflow Diagram



Appendix C: Grid Impact Results (DLR vs SLR)



The results show that applying DLR increased the allowable branch ratings for all three selected 138 kV lines. As a result, branch loading percentages decreased consistently, confirming improved transmission utilization under identical power-flow conditions.

Appendix D: N-1 Contingency Result

The following table summarizes the monitored-line response under pre-outage and post-outage conditions.

Case	Condition	Loading
Base	Pre-Outage	74.84
DLR	Pre-Outage	57.57
Base	Post-Outage	Not converged
DLR	Post-Outage	Not converged

The contingency analysis suggests that the DLR case provided improved operating margin prior to loss of convergence under stressed system conditions.

This indicates improved operational flexibility and stronger thermal margin under outage scenarios.

Appendix E: Key Code Snippets

The following code snippets represent the core implementation steps used in this project for Dynamic Line Rating (DLR) integration and contingency-based power system analysis.

E.1: Dynamic Ampacity Selection from Logged Data

This snippet selects the highest dynamic ampacity value from the experimental dataset recorded during environmental testing.

```
df = pd.read_excel(EXCEL_FILE)
best_row = df.sort_values("dynamic_ampacity_A", ascending=False).iloc[0]
dyn_ampacity_A = float(best_row["dynamic_ampacity_A"])
```

E.2: Conversion of Dynamic Ampacity to MVA

The measured dynamic ampacity was converted into an equivalent three-phase transmission line rating in MVA before being applied in PSSE.

```
dyn_mva = math.sqrt(3) * LINE_KV * (dyn_ampacity_A / 1000.0)
new_rate = min(dyn_mva, base_res["ratea"] * 1.30)
```

E.3: Updating Transmission Line Rating in PSSE

This snippet applies the new Dynamic Line Rating (DLR) to the selected transmission line inside the PSSE simulation environment.


```

intgar = [_i, _i, _i, _i, _i, _i]
realar = [r, x, charg, new_rate, new_rate, new_rate]
psspy.branch_chng(FROM_BUS, TO_BUS, CKT_ID, intgar, realar)

```

E.4: Solving the Updated Power Flow Case

After applying the new line rating, the updated power system case was solved to evaluate the impact of DLR on line loading.

```

psspy.fnsf([0, 0, 0, 1, 1, 0, 99, 0])

```

E.5: Simulating an N-1 Contingency

This code removes a selected transmission line from service to test the effect of a single line outage on the monitored branch.

```

intgar = [0, _i, _i, _i, _i, _i]
realar = [r, x, charg, _f, _f, _f]
psspy.branch_chng(OUT_FROM, OUT_TO, OUT_CKT, intgar, realar)

```

E.6: Extracting Line Loading Results

This snippet reads the active power, reactive power, and percentage loading of the monitored line before and after applying DLR and contingency conditions.

```

ierr, p = psspy.brnmsc(FROM_BUS, TO_BUS, CKT_ID, "P")
ierr, q = psspy.brnmsc(FROM_BUS, TO_BUS, CKT_ID, "Q")
ierr, pct = psspy.brnmsc(FROM_BUS, TO_BUS, CKT_ID, "PCTMVA")

```